

Global Startup Ecosystem Analysis

Project Change Log

OVERVIEW

This change log documents the chronological modifications, data cleaning steps, and validation checks performed throughout the analysis of the global startup dataset. All data cleaning adjustments and transformations were verified and tested using SQL queries.

1. INITIAL DATA INGESTION & INSPECTION

- Raw Data Import: Imported raw CSV files (startups.csv, funding_rounds.csv, investors.csv, quarterly_summary.csv) sourced from Kaggle under the CC0 Public Domain license.
- Schema Mapping: Established relational structures across tables using foreign keys such as startup_id and round_id.

2. DATA CLEANING & TRANSFORMATION PHASE

- Missing Value Handling: Identified and removed rows containing critical missing values across essential metrics to prevent analytical skew.
- Text & Field Standardization: Cleaned text formatting issues in regional dimensions (e.g., country, city, sector names) to ensure accurate relational grouping.
- Data Type Enforcement: Verified numeric constraints on financial columns (such as amount_million_usd, post_money_valuation_million_usd, and num_investors) to support proper mathematical aggregation.

3. SQL VALIDATION & VERIFICATION PHASE

- Relational Integrity Checks: Executed multi-table SQL JOIN statements to verify that foreign key relationships between startups and funding rounds remained intact without data corruption.
- Aggregation Testing: Tested GROUP BY queries across geographic regions and industry sectors to ensure accurate counts and sums.
- Outlier Verification: Ran MAX() valuation queries to isolate and cross-verify high-valuation data points (such as CruxWare) against raw table rows to guarantee analytical accuracy.

4. VISUALIZATION & REPORTING PHASE

- Tableau Integration: Connected the cleaned, SQL-validated tables to Tableau Public to generate proportional treemaps and performance bar charts.
- Repository Documentation: Structured the GitHub repository by organizing SQL scripts, dashboard screenshots, and the project README.

SUMMARY

All data cleaning steps in this project were validated using SQL queries to confirm accuracy before any visualization or reporting work took place, ensuring the final dashboards and findings are built on a verified dataset.

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